

TASK - 2

1. If-Else Statements

These let your program make decisions, like choosing what to wear based on the weather. If it's raining, grab an umbrella; else, wear sunglasses. They check a condition and run one block of code if true, another if false.

2. Comparison Operators

These are the questions your code asks to make those decisions: `==` for "equals?", `!=` for "not equals?", `>` for "greater than?", `<` for "less than?", and so on with `>=` or `<=`. They're like quick checks before deciding what to do next.

3. For Loops

Picture repeating a task a set number of times, say printing "Hello" five times automatically. For loops handle that by iterating over a range or list, making repetitive work a breeze without copying code.

4. While Loops

These keep going until a condition changes, like "keep eating snacks while you're hungry." They check the condition before each round and stop when it's false, great for situations where you don't know the exact repeat count upfront.

5. Functions

Think of functions as reusable recipes—you write the steps once, name it, and call it whenever needed, maybe passing in ingredients (parameters). This keeps code clean, saves time, and avoids repetition.

6. Basic Error Handling

When code hits a snag, like dividing by zero, error handling (try-except in Python) catches it gracefully instead of crashing. It's like having a safety net: try the risky move, and if it fails, do something else or show a helpful message.

7. Time messages

```
time = int(input("Enter the hour in 24-hour format: "))
if time < 12:
    print("Good morning!")
elif time < 17:
    print("Good afternoon!")
elif time < 21:
    print("Good evening!")
else:
    print("Good night!")
```

8. Guessing and loops

```
secret_number = 7
guess = int(input("Guess the number: "))
if guess == secret_number:
    print("Correct! You guessed it.")
else:
    print("Wrong guess. Try again!")
```

9. Use For Loop to print 1 to 100

```
for i in range(1, 101):
    print(i)
```

10. Functions and calculator

```
def area_rectangle(length, width):
    return length * width

length = float(input("Enter length: "))
width = float(input("Enter width: "))
area = area_rectangle(length, width)
print("Area of rectangle is:", area)
```

11. Calculator

```
def add(a, b):
    return a + b
```

```

def subtract(a, b):
    return a - b
def multiply(a, b):
    return a * b
def divide(a, b):
    if b == 0:
        return "Cannot divide by zero"
    return a / b
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))
operation = input("Enter operation (+, -, *, /): ")
if operation == "+":
    print("Result:", add(num1, num2))
elif operation == "-":
    print("Result:", subtract(num1, num2))
elif operation == "*":
    print("Result:", multiply(num1, num2))
elif operation == "/":
    print("Result:", divide(num1, num2))
else:
    print("Invalid operation")

```

12. Grading system

```

marks = int(input("Enter your marks: "))
if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 50:
    print("Grade C")
else:
    print("Fail")

```

13. Age categories

```

age = int(input("Enter your age: "))
if age < 13:
    print("Child")
elif age < 20:
    print("Teenager")
elif age < 60:
    print("Adult")
else:
    print("Senior Citizen")

```

```
PS C:\Users\91707\Desktop\personal intro project> git add .
PS C:\Users\91707\Desktop\personal intro project> git commit -m "Initial commit"
[master (root-commit) 642f1ea] Initial commit
 4 files changed, 47 insertions(+)
 create mode 100644 README.md
 create mode 100644 peronal intro.py
 create mode 100644 requirements.txt
 create mode 100644 screenshort.png
PS C:\Users\91707\Desktop\personal intro project> git remote add origin http
s://github.com/prinu916/Internship-Task-for-The-Developer-Sarena.git
PS C:\Users\91707\Desktop\personal intro project> git branch -M main
PS C:\Users\91707\Desktop\personal intro project> git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 43.45 KiB | 10.86 MiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/prinu916/Internship-Task-for-The-Developer-Sarena.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\91707\Desktop\personal intro project>
```

=== Student Grade Calculator ===

Enter student name: Priyanshu

Enter marks (0-100): 95

 RESULT FOR PRIYANSHU:

Marks: 95/100

Grade: A

Message: Excellent work! You should be proud of yourself! 🌟

=== Student Grade Calculator ===

Enter student name: Aman

Enter marks (0-100): 85

 RESULT FOR AMAN:

Marks: 85/100

Grade: B

Message: Very Good! Keep it up! 👍

```
=== Student Grade Calculator ===
```

```
Enter student name: Maya
```

```
Enter marks (0-100): 74
```

```
📁 RESULT FOR MAYA:
```

```
Marks: 74/100
```

```
Grade: C
```

```
Message: Good job! You are doing well, keep improving! 😊
```

```
=== Student Grade Calculator ===
```

```
Enter student name: Chanchal
```

```
Enter marks (0-100): 62
```

```
📁 RESULT FOR CHANCHAL:
```

```
Marks: 62/100
```

```
Grade: D
```

```
Message: Nice effort! Keep practicing and you will get even better! 🍷
```

```
=== Student Grade Calculator ===
```

```
Enter student name: Mann
```

```
Enter marks (0-100): 40
```

```
📁 RESULT FOR MANN:
```

```
Marks: 40/100
```

```
Grade: F
```

```
Message: Don't give up. Every expert started as a beginner. Keep trying! ❤️
```

```
=== Student Grade Calculator ===
```

```
Enter student name: nmc
```

```
Enter marks (0-100): cdjn
```

```
Invalid input. Please enter a whole number only.
```

```
Enter marks (0-100): xc
```

```
Invalid input. Please enter a whole number only.
```

```
Enter marks (0-100): 105
```

```
Invalid input. Marks must be between 0 and 100.
```

```
Enter marks (0-100): 105
```

```
Invalid input. Marks must be between 0 and 100.
```

```
Enter marks (0-100):
```